THREE EQUIVALENT CHARACTERISATIONS OF SEMINORMAL SCHEMES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file morphisms.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The step '(1) iff (2) is clear from Lemma lemma-seminormalization' is a citation of the in-chapter lemma which constructs the seminormalization and states its universal property, i.e. the defining property of the object being characterised, while the substantive implication (2) implies (3) is proved inside the counted span; accepted, not treated as an omitted statement item.

Allowed standard prerequisites (editorial).

- The seminormalization is universal among seminormal schemes over the scheme. Lemma \ref{lemma-seminormalization}, cited at proof lines 12312-12318 and used again at proof lines 12318-12322. Gives the equivalence of (1) and (2) and produces the factorisation used for (2) implies (3).
- A section of a separated morphism is a closed immersion. Schemes, Lemma \ref{schemes-lemma-section-immersion} and Lemma \ref{lemma-universally-injective-separated}, cited at proof lines 12323-12326. Shows the first arrow of the factorisation is a closed immersion.
- A closed immersion which is bijective on points onto a reduced scheme is an isomorphism. used at proof lines 12326-12327. Finishes the proof of (2) implies (3).

Source reference. Original lemma 0H3G: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 21316–21316. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 0EUS: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 18562–18562. Navigation pointer to a statement the

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counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 01KT: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2062–2062. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 05VE: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7548–7548. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let X be a scheme. The following are equivalent*

(1) *X is seminormal,*
(2) *X is equal to its own seminormalization, i.e., the morphism $X^{sn} \to X$ is an isomorphism,*
(3) *if $\pi : Y \to X$ is a universal homeomorphism inducing isomorphisms on residue fields with Y reduced, then π is an isomorphism.*

Proof. The equivalence of (1) and (2) is clear from Lemma 0EUS. If (3) holds, then $X^{sn} \to X$ is an isomorphism and we see that (2) holds.

Assume (2) holds and let $\pi : Y \to X$ be a universal homeomorphism inducing isomorphisms on residue fields with Y reduced. Then there exists a factorization $X \to Y \to X$ of id_X by Lemma 0EUS. Then $X \to Y$ is a closed immersion (by Schemes, Lemma 01KT and the fact that π is separated for example by Lemma 05VE). Since $X \to Y$ is also a bijection on points, the reducedness of Y shows that it has to be an isomorphism. This finishes the proof. □